

Troubleshooting Guide – Smart Irrigation System

This guide helps developers, technicians, and operators identify and fix common issues.

1. Backend Issues

Problem	Possible Cause	Solution
Backend not starting	Missing <code>.env</code> variables	Check <code>.env</code> file and restart backend
Python dependency errors	<code>requirements.txt</code> outdated	Run <code>pip3 install -r requirements.txt</code>
API returns 500	Rule engine crash or DB error	Check logs: <code>docker logs irrigation-backend</code>
Port 8000 already in use	Another app running	Change port in <code>docker-compose.yml</code> or stop conflicting service

2. Database Issues

Problem	Possible Cause	Solution
Cannot connect to DB	Service not running	<code>docker-compose up -d db</code>
Connection refused	Wrong <code>DB_URL</code> in <code>.env</code>	Correct URL format: <code>postgresql://user:pass@db:5432/dbname</code>
Data missing	Migration not run	Ensure tables created, run DB setup scripts

3. Rule Engine Issues

Problem	Possible Cause	Solution
Irrigation never turns ON	Soil thresholds too high	Adjust <code>LOWER_LIMIT</code> in <code>rule_engine.py</code>
Irrigation always ON	Thresholds too low	Adjust <code>UPPER_LIMIT</code> in <code>rule_engine.py</code>
ET calculation incorrect	PyETo misconfigured	Validate inputs, check PyETo version
Sensor data missing	Gateway not sending	Check ESP32 connectivity

4. Sensor & IoT Issues

Problem	Possible Cause	Solution
No data received	ESP32 not powered	Verify power supply and network

Problem	Possible Cause	Solution
Wrong sensor values	Calibration issue	Recalibrate sensor or replace
Rain sensor not detected	Wiring issue	Check connections, verify GPIO

5. Swagger UI / API Docs Issues

Problem	Possible Cause	Solution
Swagger not loading	Volume path incorrect	Check <code>docker-compose.yml</code> mounts
Wrong endpoints	Outdated <code>openapi.yaml</code>	Update file and restart container
404 error	Wrong URL	Use <code>http://<server-ip>:8081/</code>

6. Docker Issues

Problem	Possible Cause	Solution
Container crashes	Missing dependencies	Rebuild: <code>docker-compose up --build -d</code>
Port conflict	Another service running	Change mapped ports in <code>docker-compose.yml</code>
Containers not starting on reboot	Docker not enabled	<code>sudo systemctl enable docker</code>

7. Logs & Debugging

- Backend: `docker logs -f irrigation-backend`
- DB: `docker logs -f irrigation-db`
- Rule Engine: check `engine.py` debug prints
- Network: `ping <ESP32-IP>` or `curl` API endpoints

8. Tips

- Always validate sensor values before sending to backend.
- Keep `.env` secure and updated.
- Use version control for `openapi.yaml` to track API changes.
- Perform back-testing for rule engine thresholds after major changes.
- Restart services in order: DB → Backend → Swagger UI.